

House Price Prediction

Exploratory Data Analysis (EDA)

- Dataset: Ames Housing Dataset
- Number of Samples: 1460
- Number of Features: 80
- Target Variable: **SalePrice**



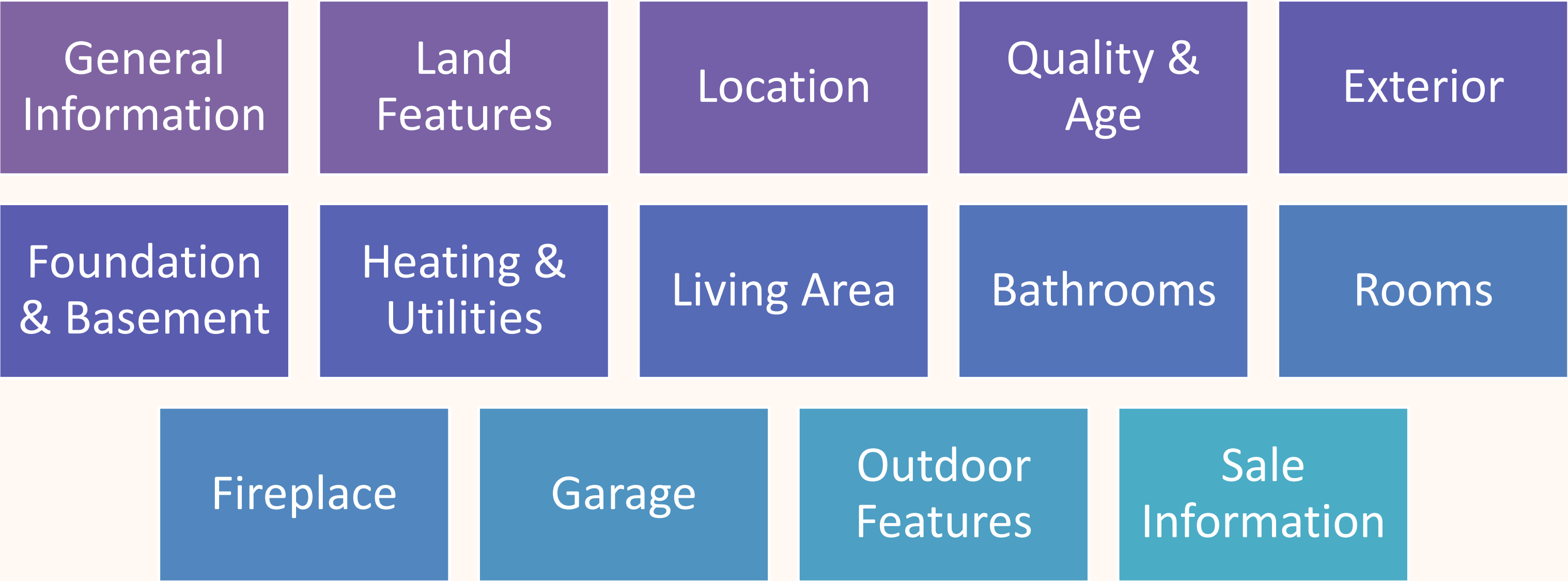
Dataset Overview



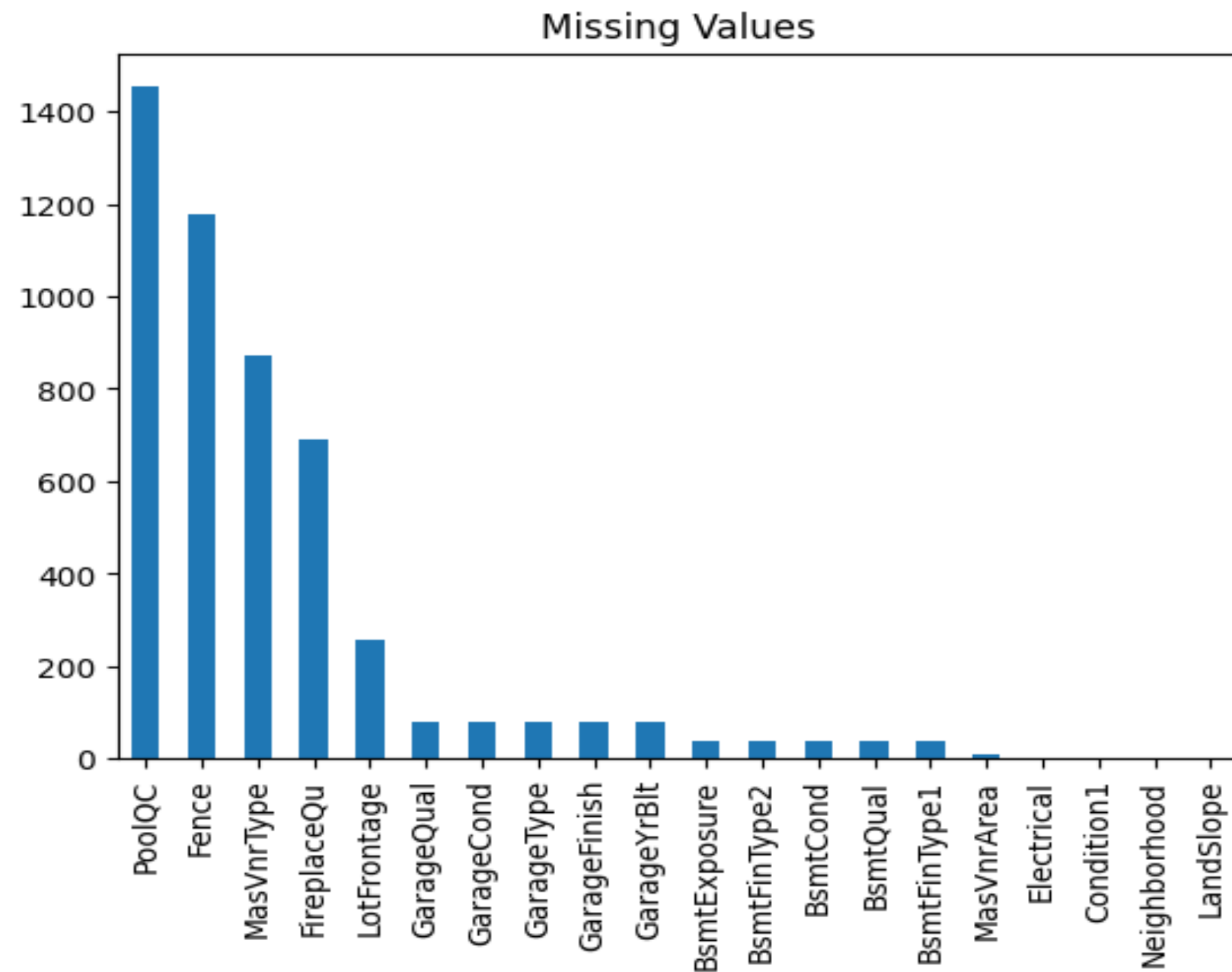
Item	Value
Rows	1460
Columns	81
Numerical	38
Categorical	43
Target	SalePrice

The dataset contains structural, geographical, and transactional information used to predict house prices.

Feature Categorization



Missing Values



Alley → 93.8%

PoolQC → 99.5%

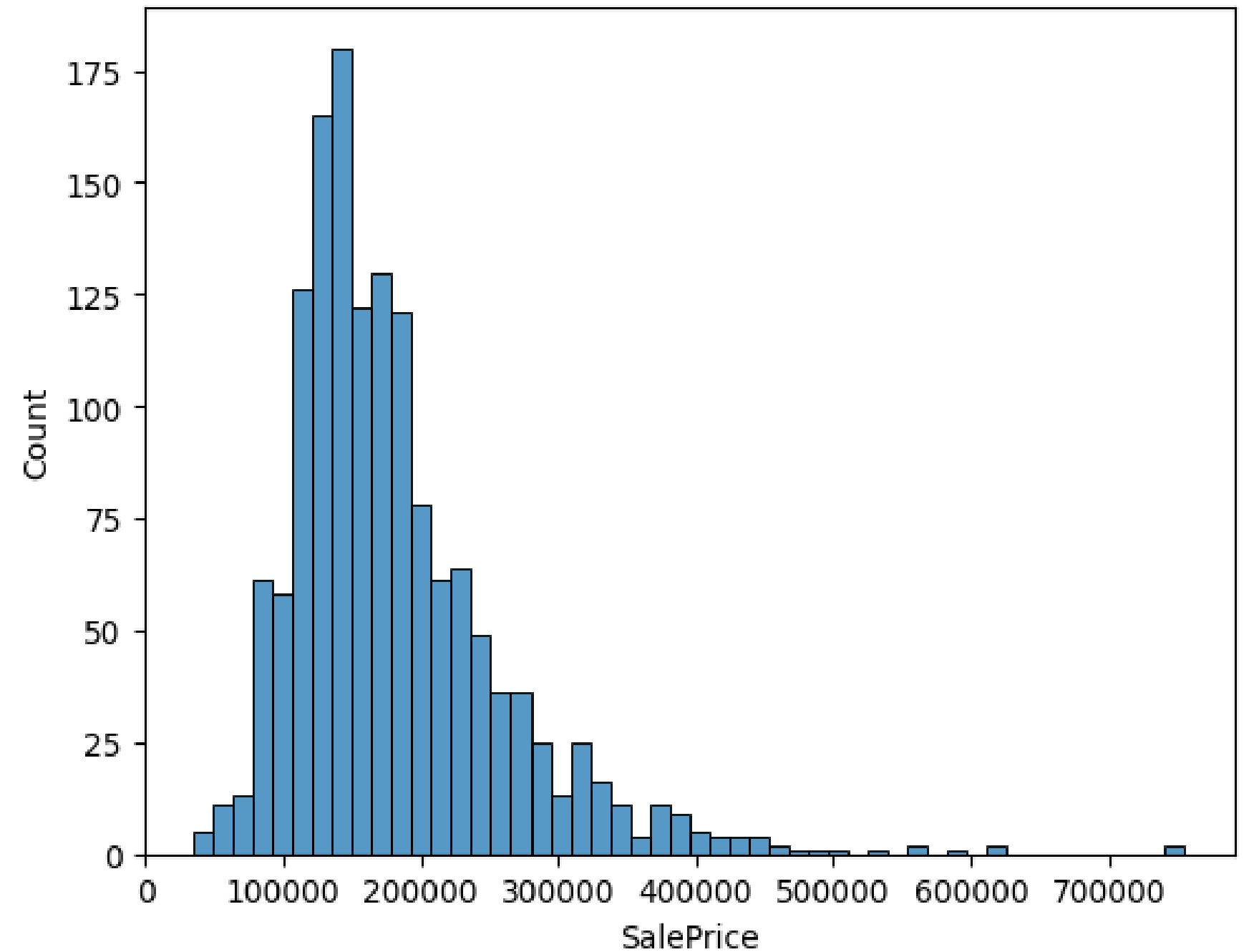
Fence → 80%

MiscFeature → 96%

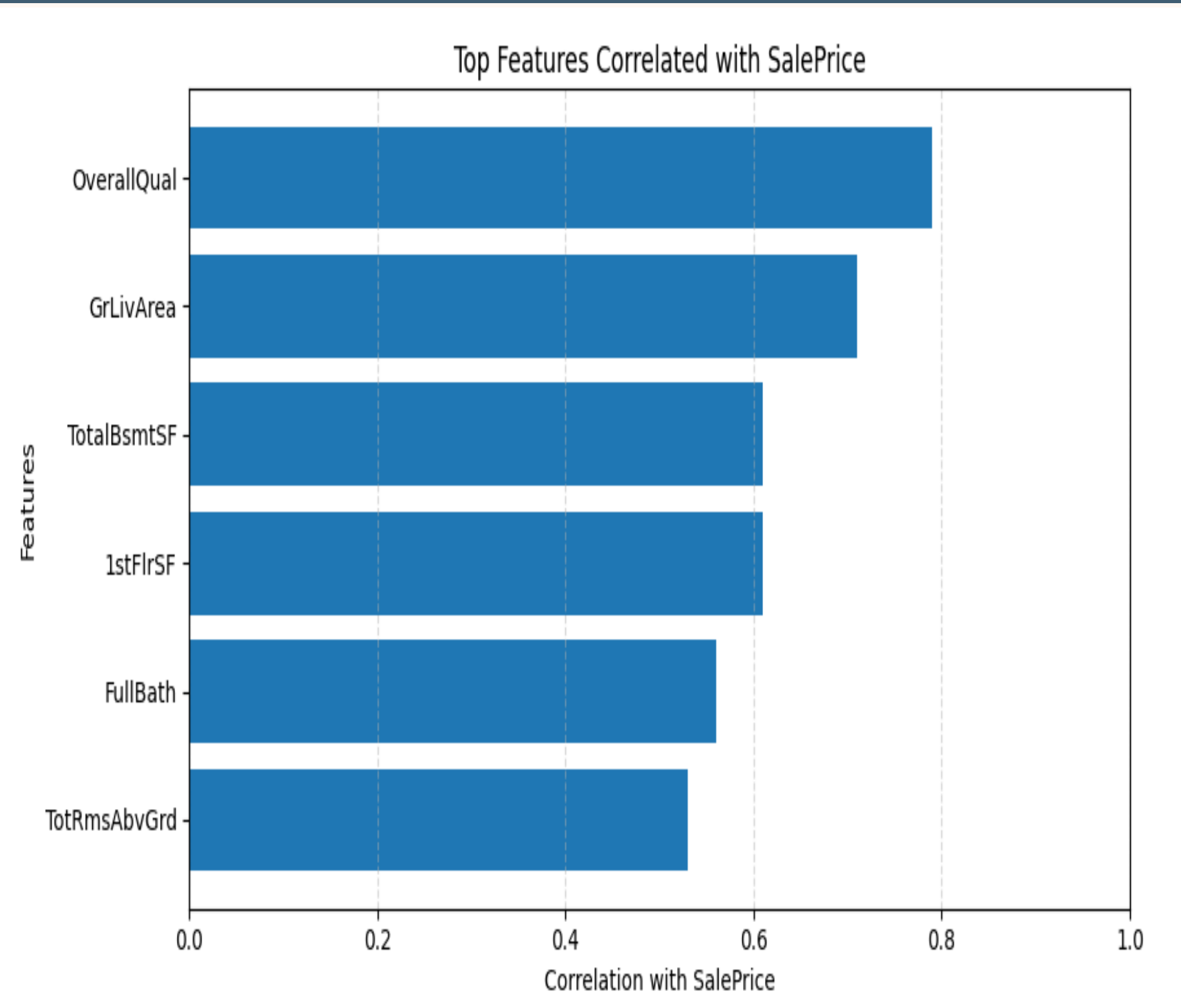
Most missing values indicate the absence of a feature rather than missing information.

Target Variable

- Right-skewed
- Presence of expensive houses (Outliers)
- Log Transformation may improve model performance

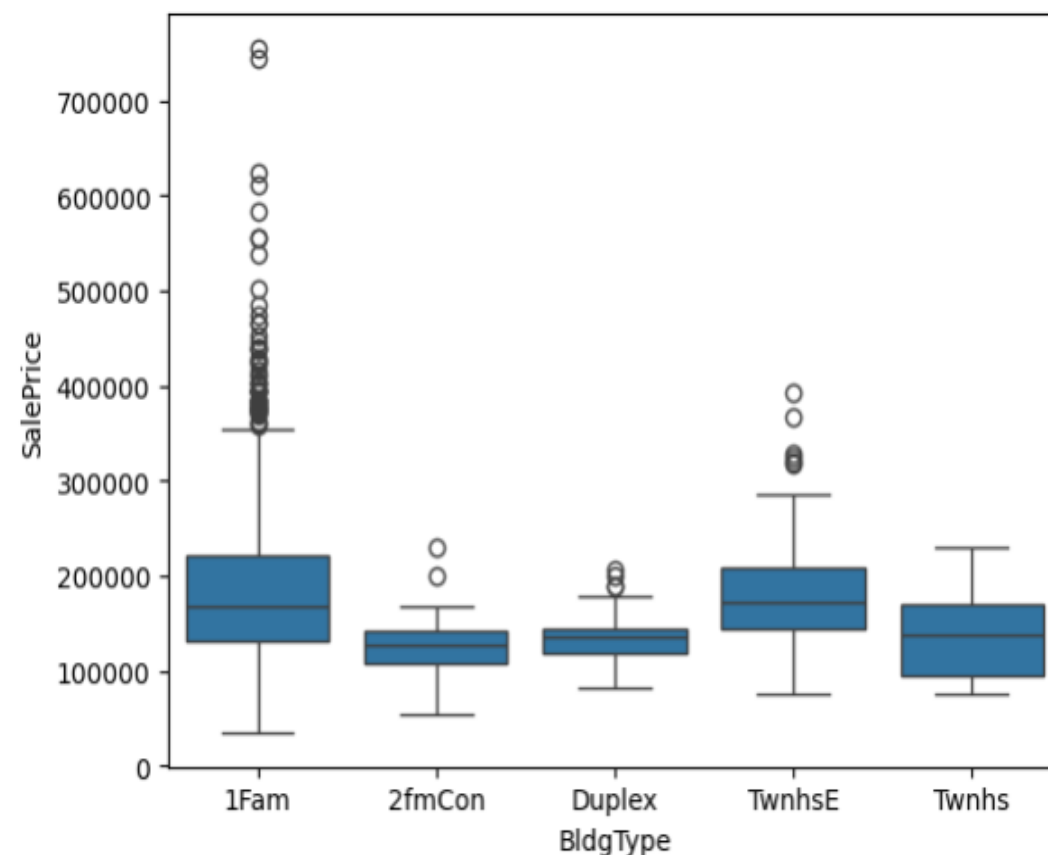


Correlation with SalePrice

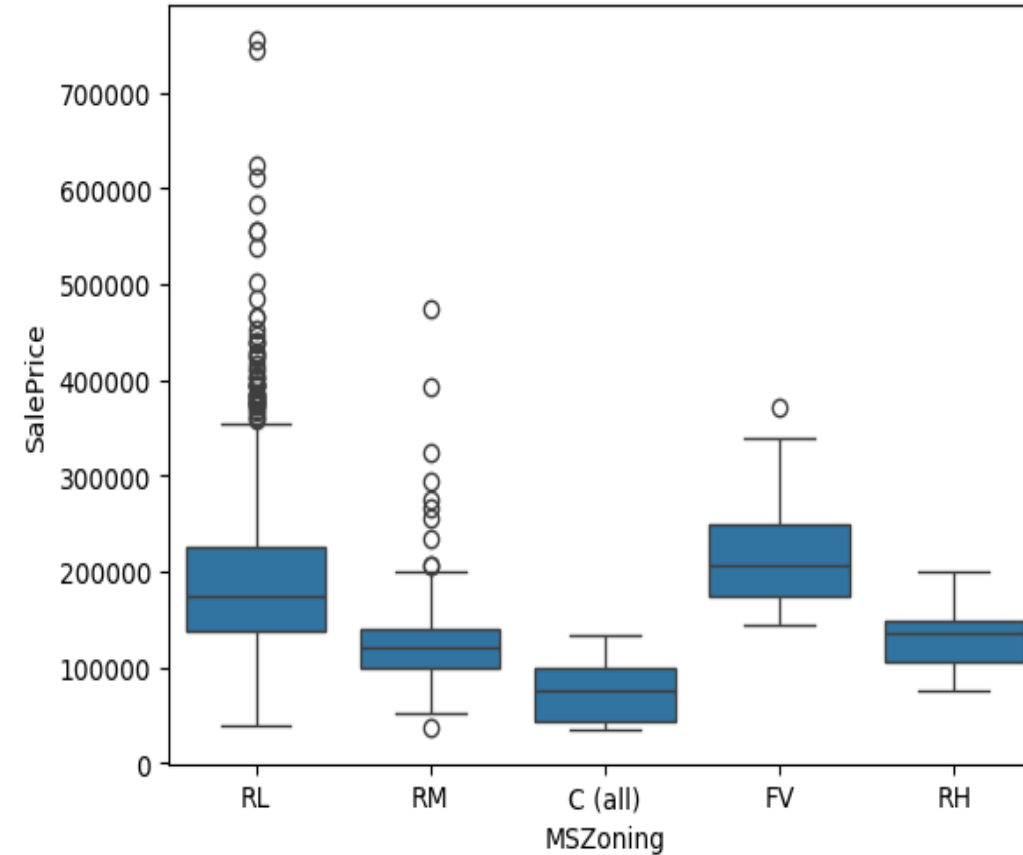


General Information

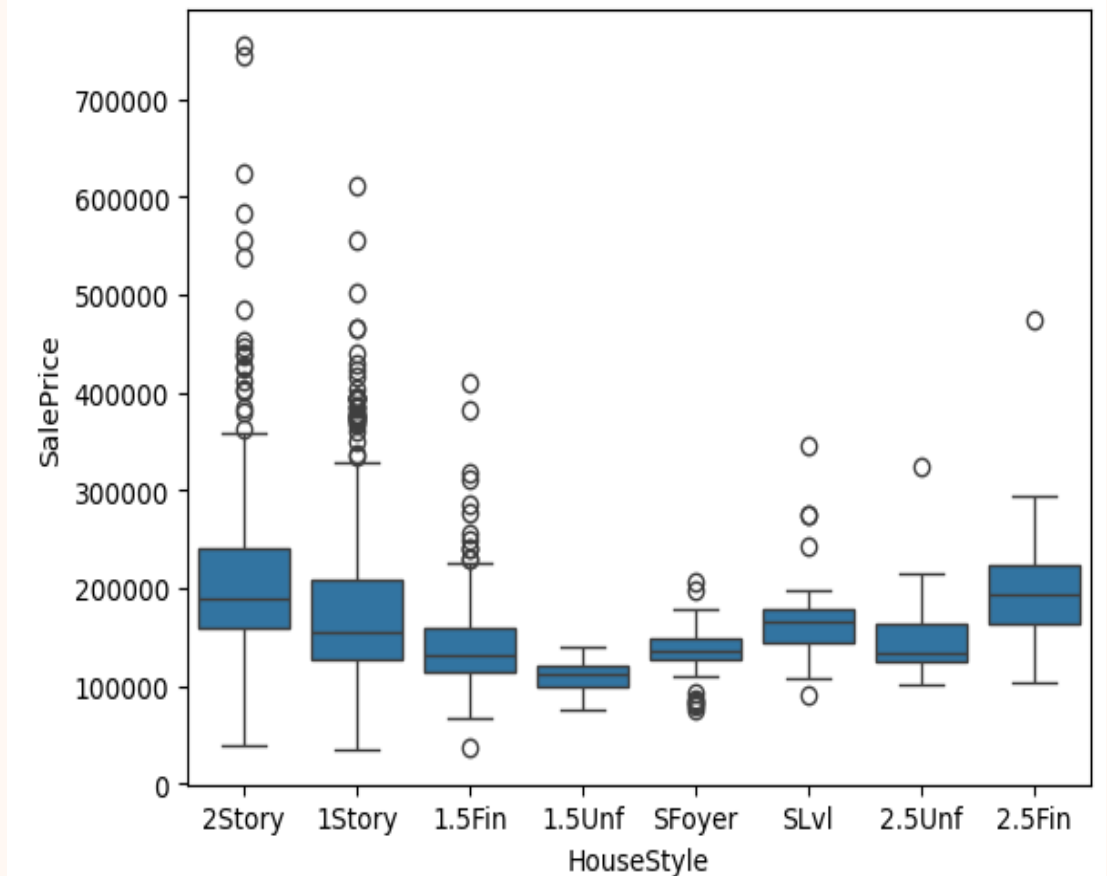
BldgType



MSZoning



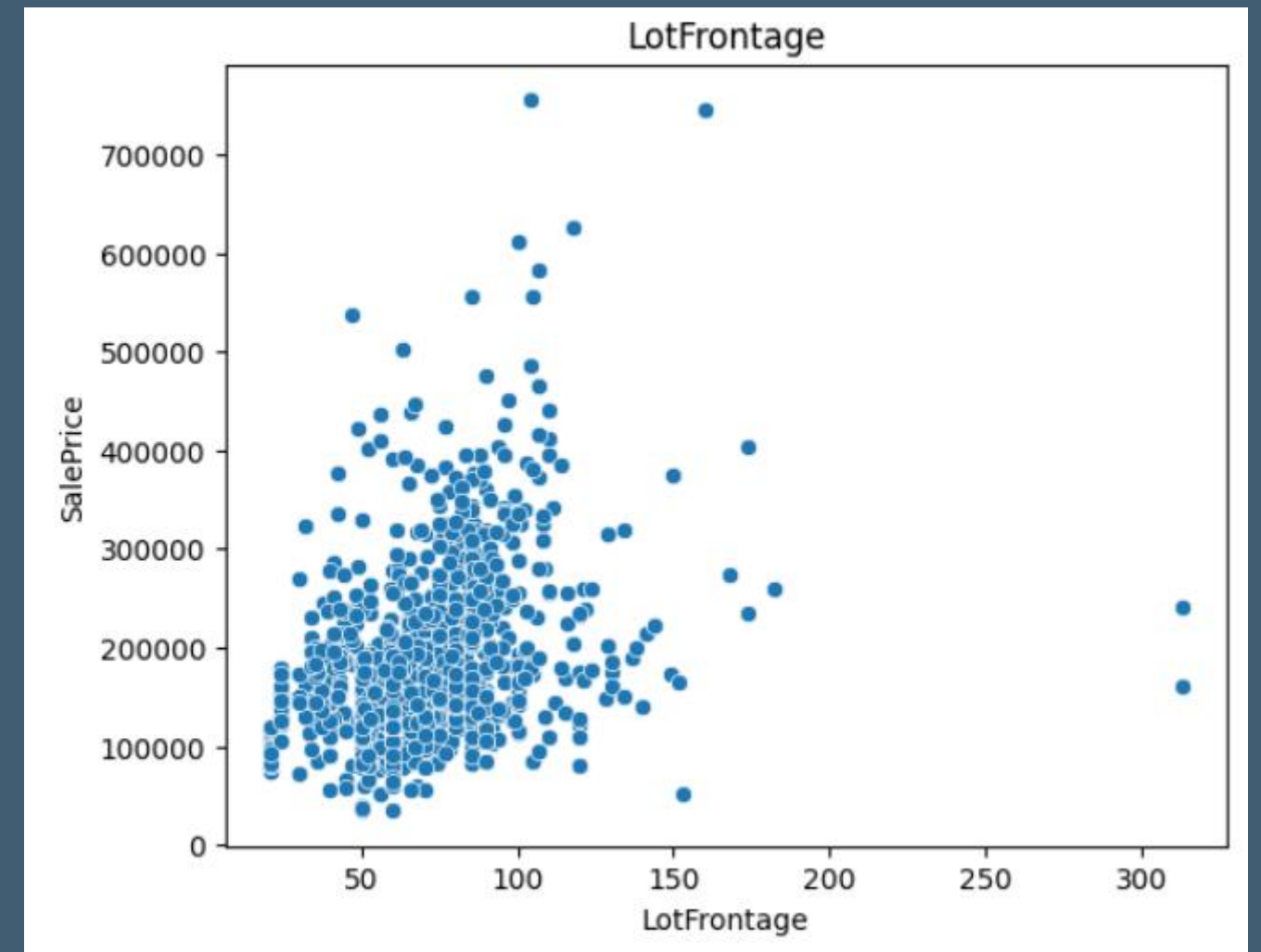
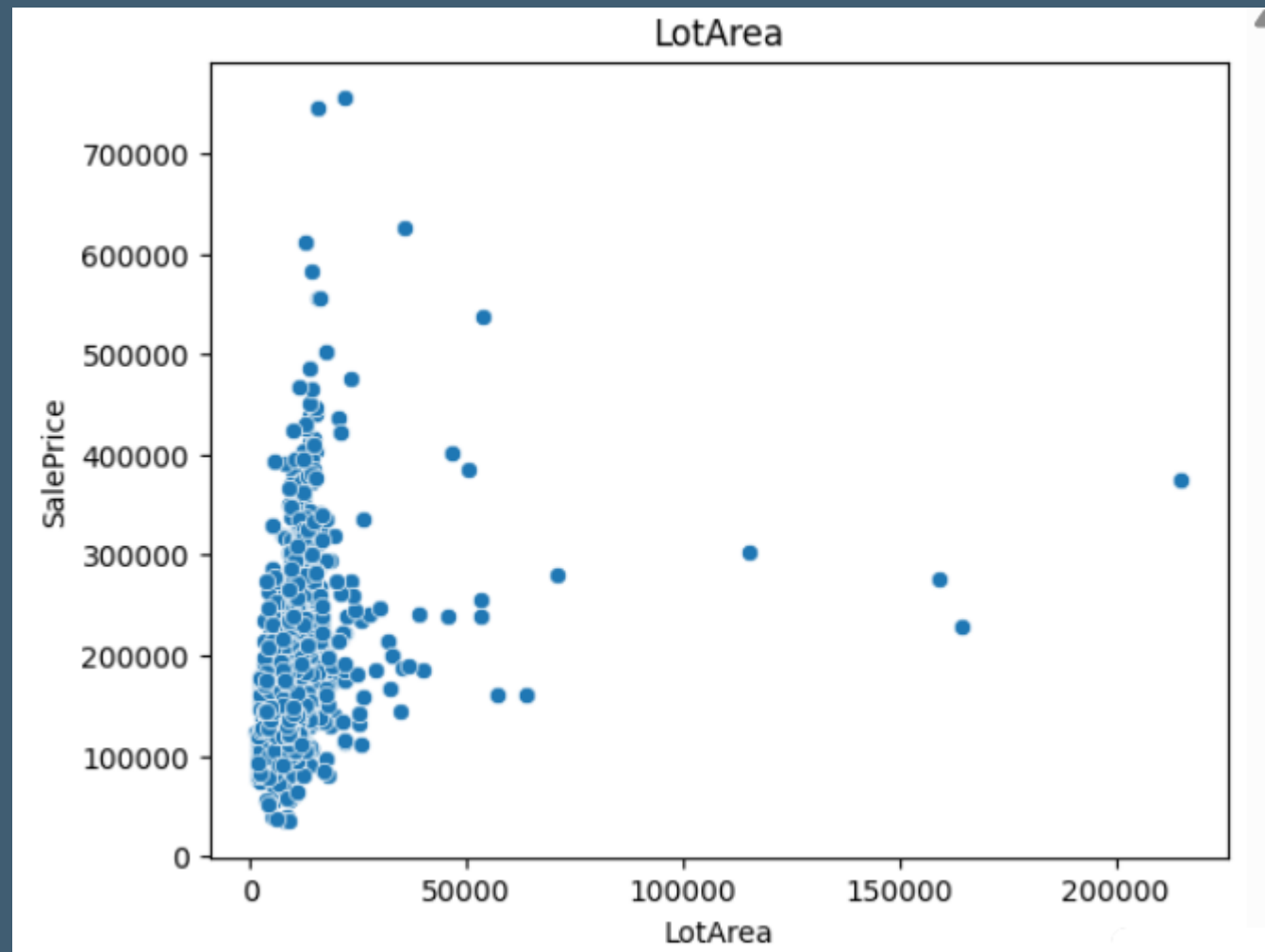
HouseStyle



Key Findings

- Residential zoning affects prices.
- House style influences property value.
- Building type shows clear price differences.

Land Features



- Larger lots generally have higher prices.
- Irregular lots tend to be more expensive.
- Steep land has limited influence.

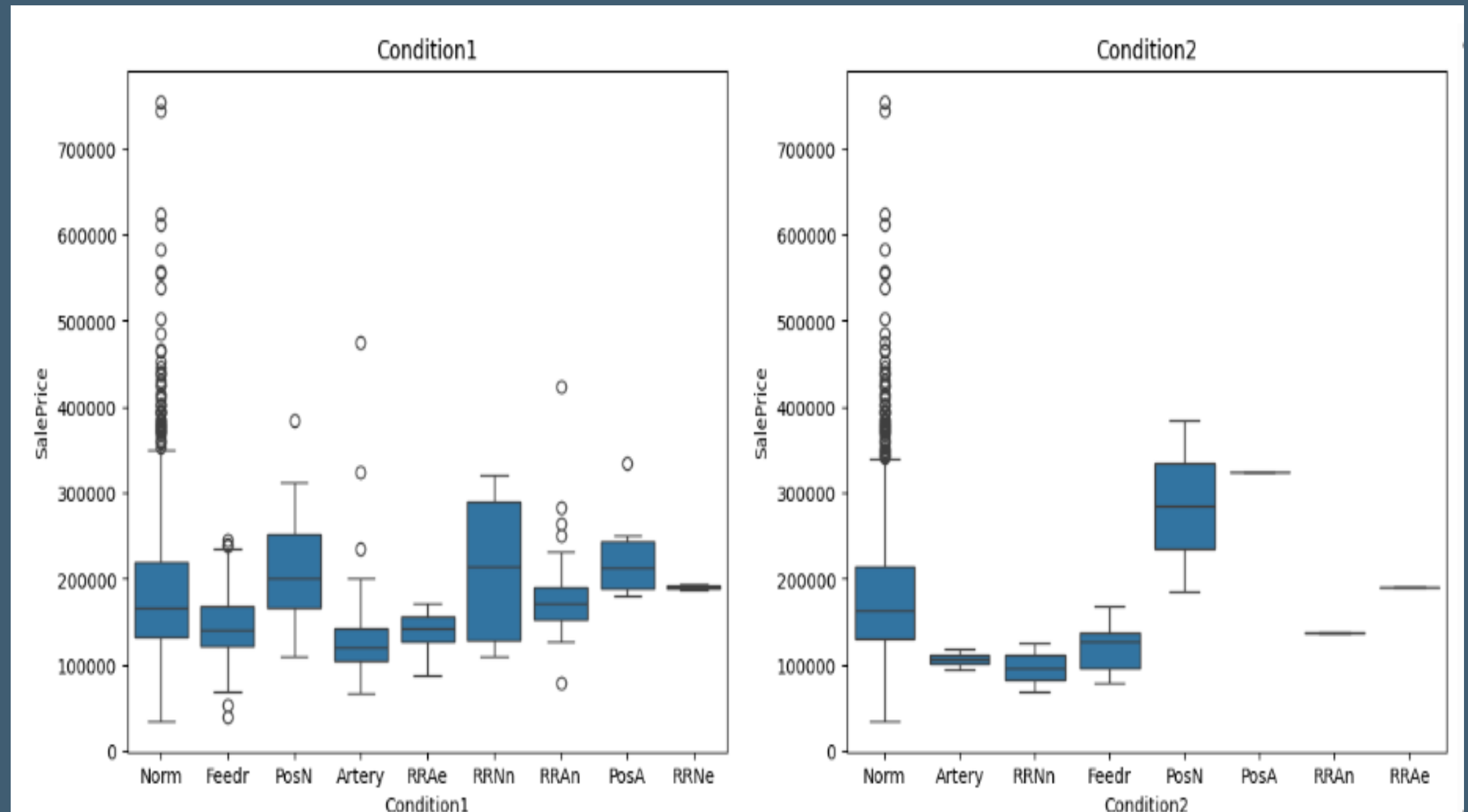
Location

Top Neighborhoods

- NoRidge
- NridgHt
- StoneBr
- Timber
- Somerst

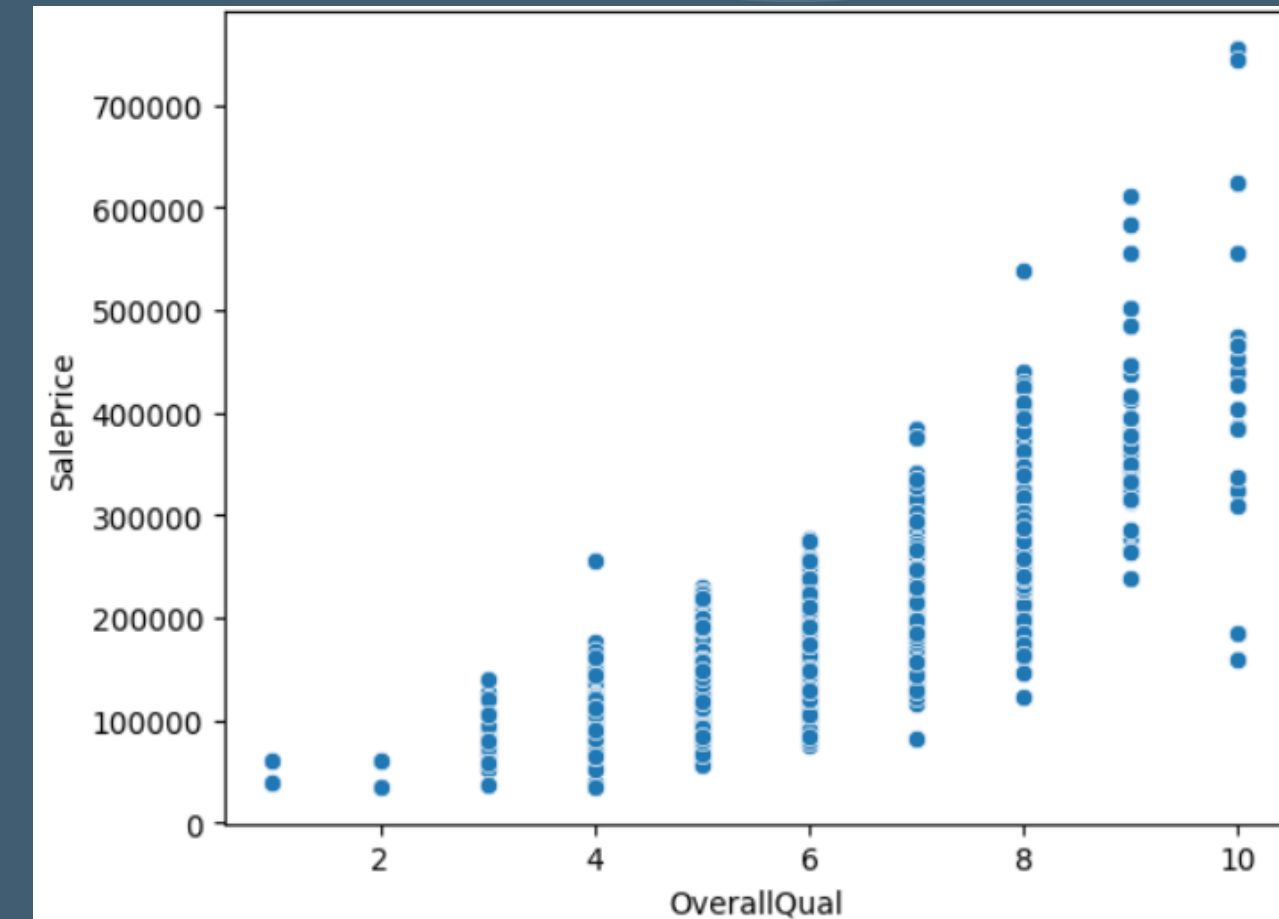
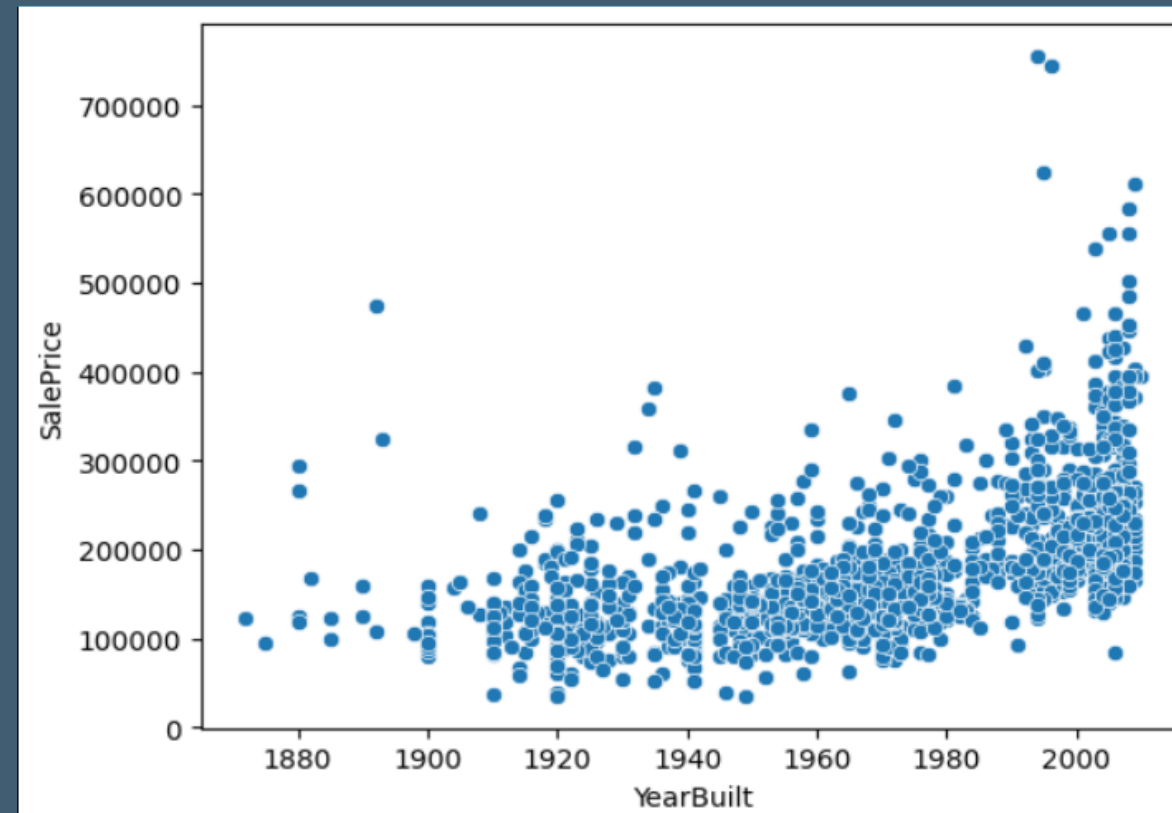
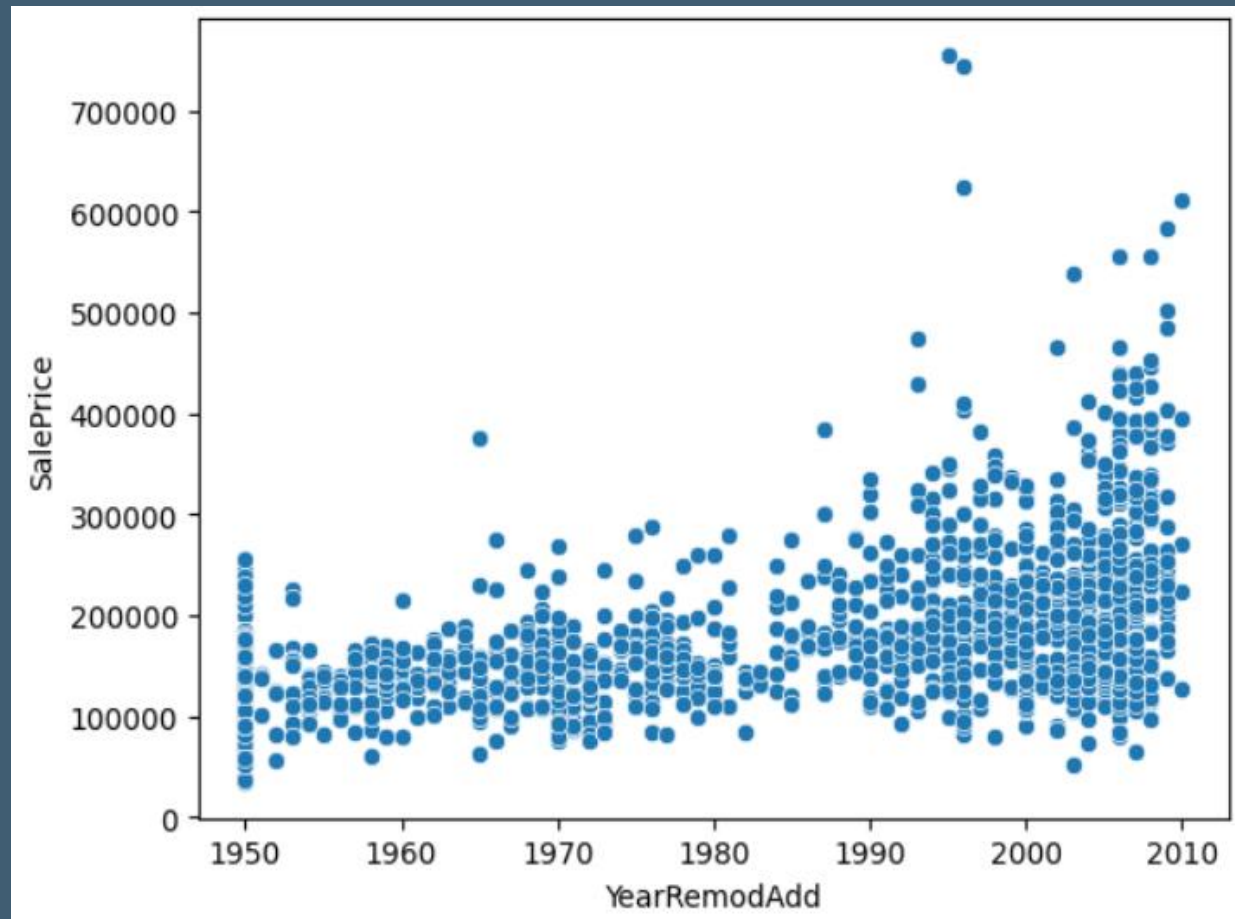
Bottom

- IDOTRR
- BrDale
- BrkSide



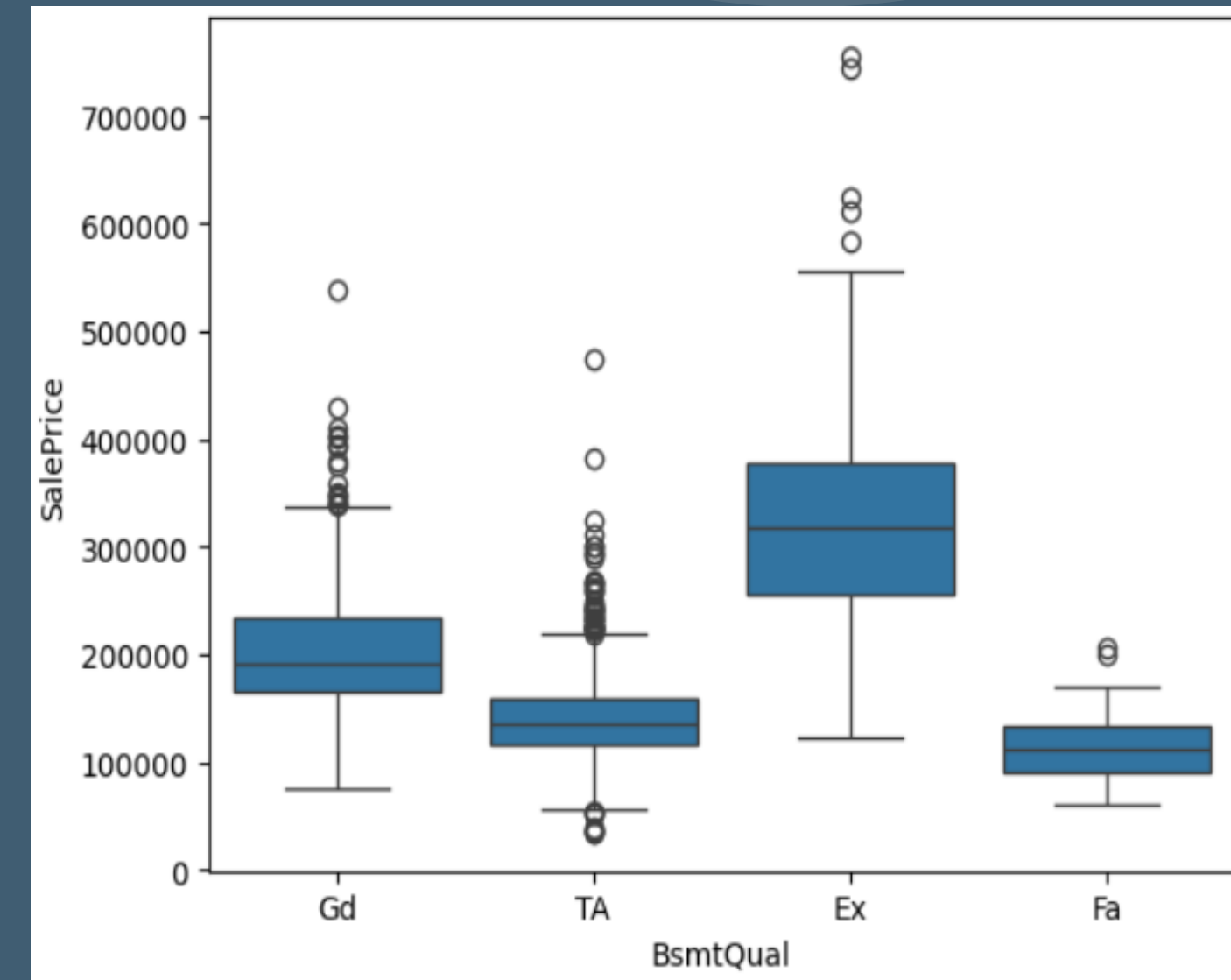
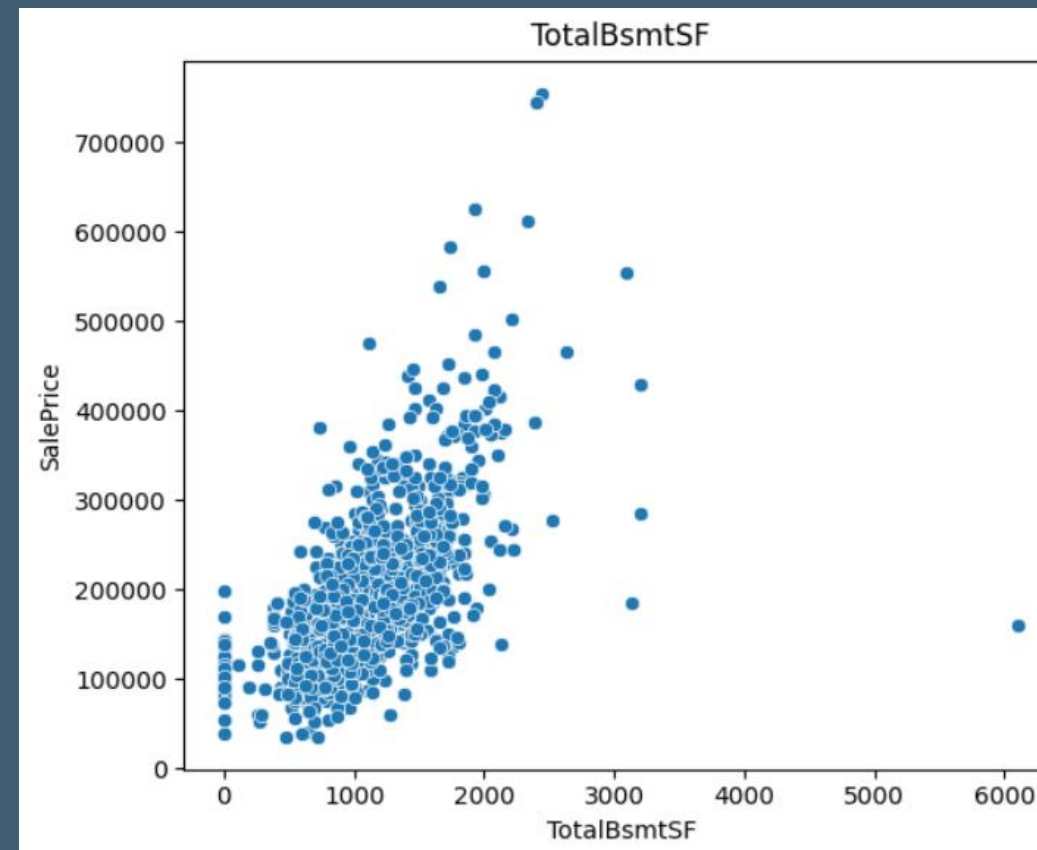
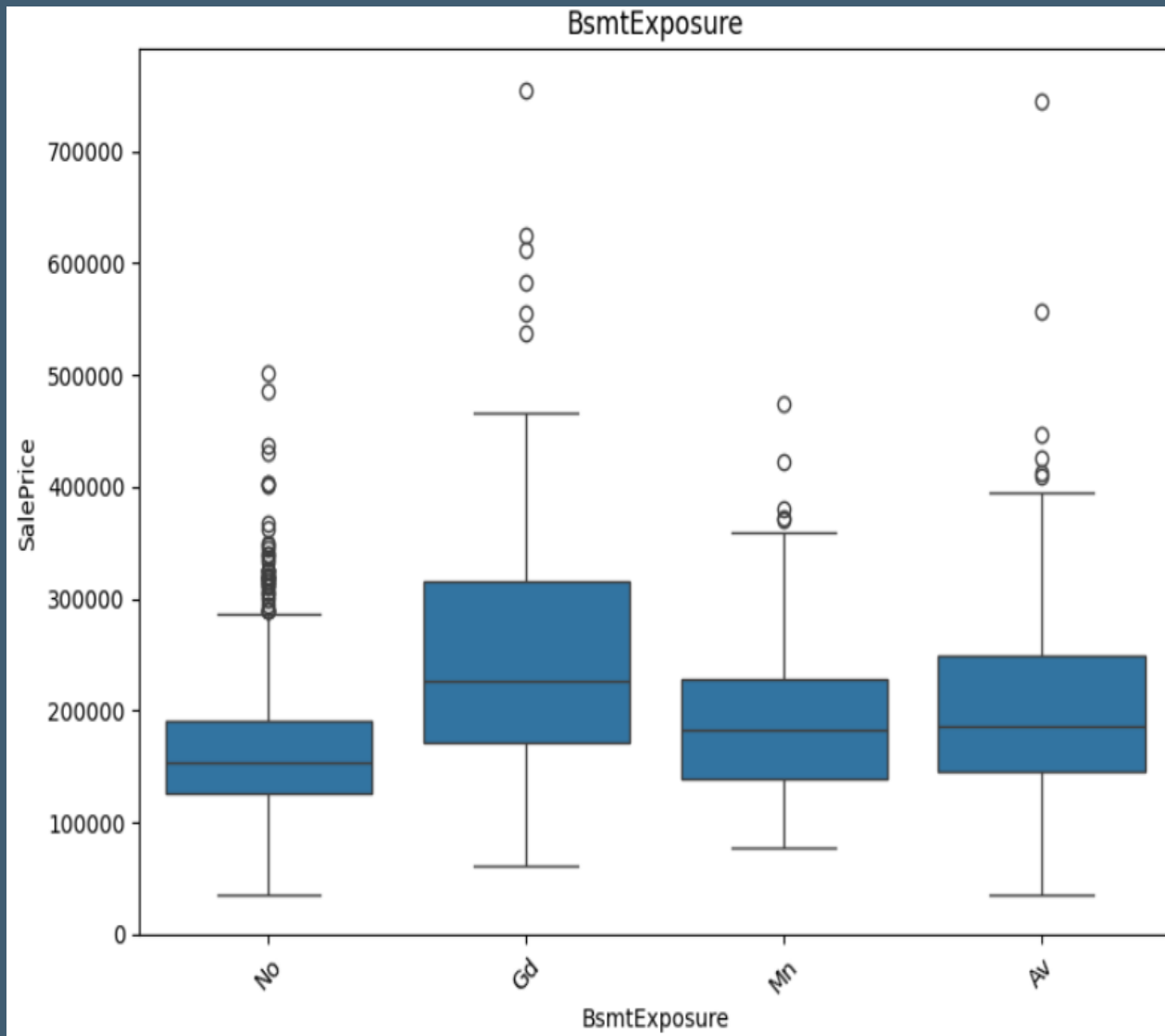
Neighborhood is one of the strongest categorical predictors.

Quality & Age



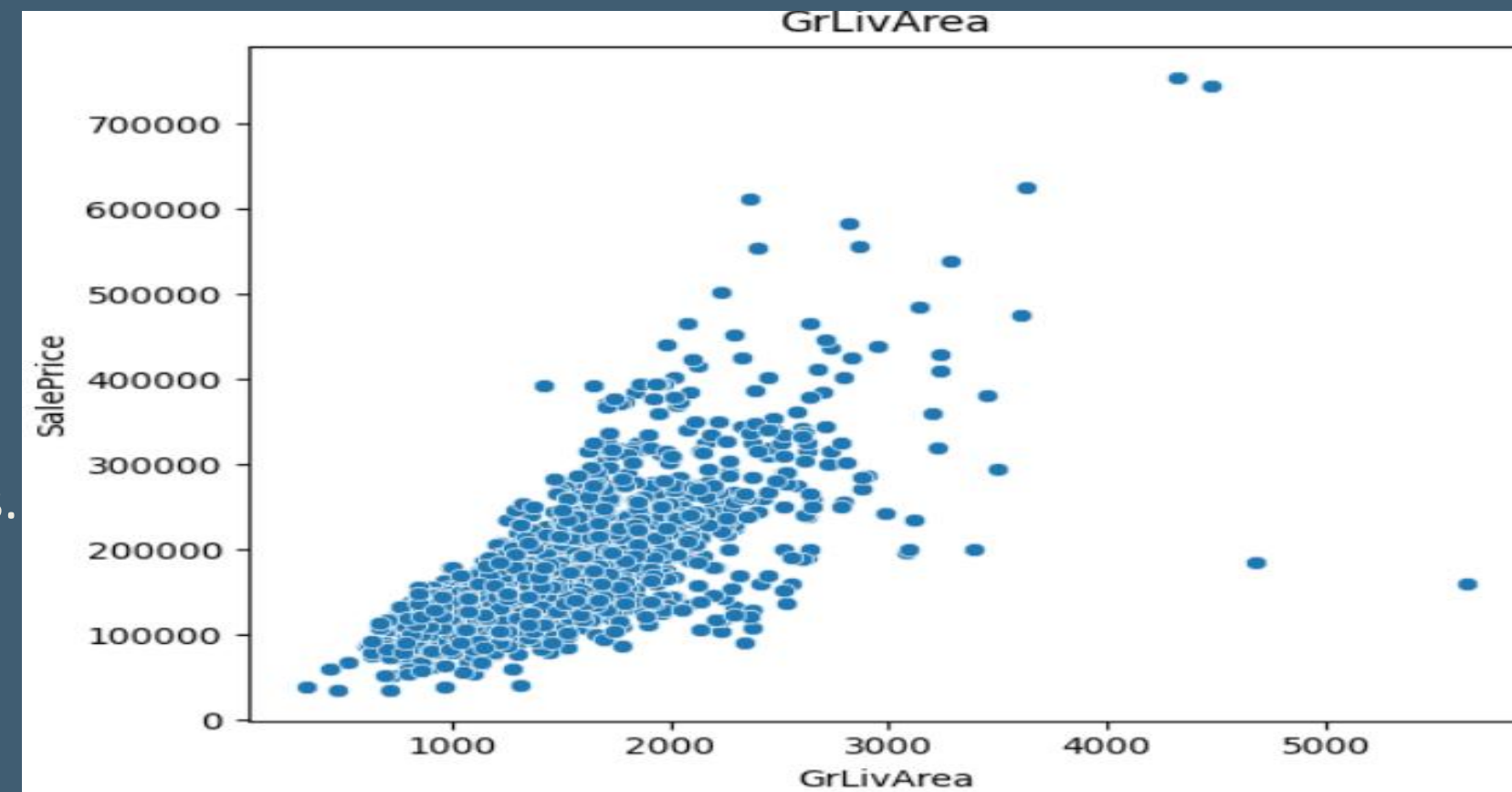
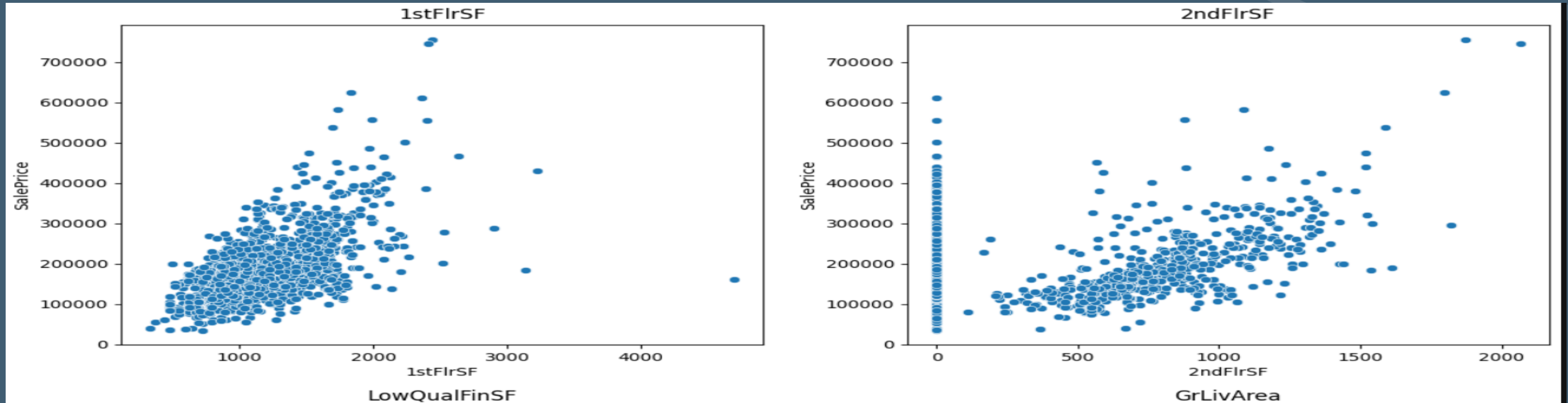
OverallQual Correlation = 0.79

Basement



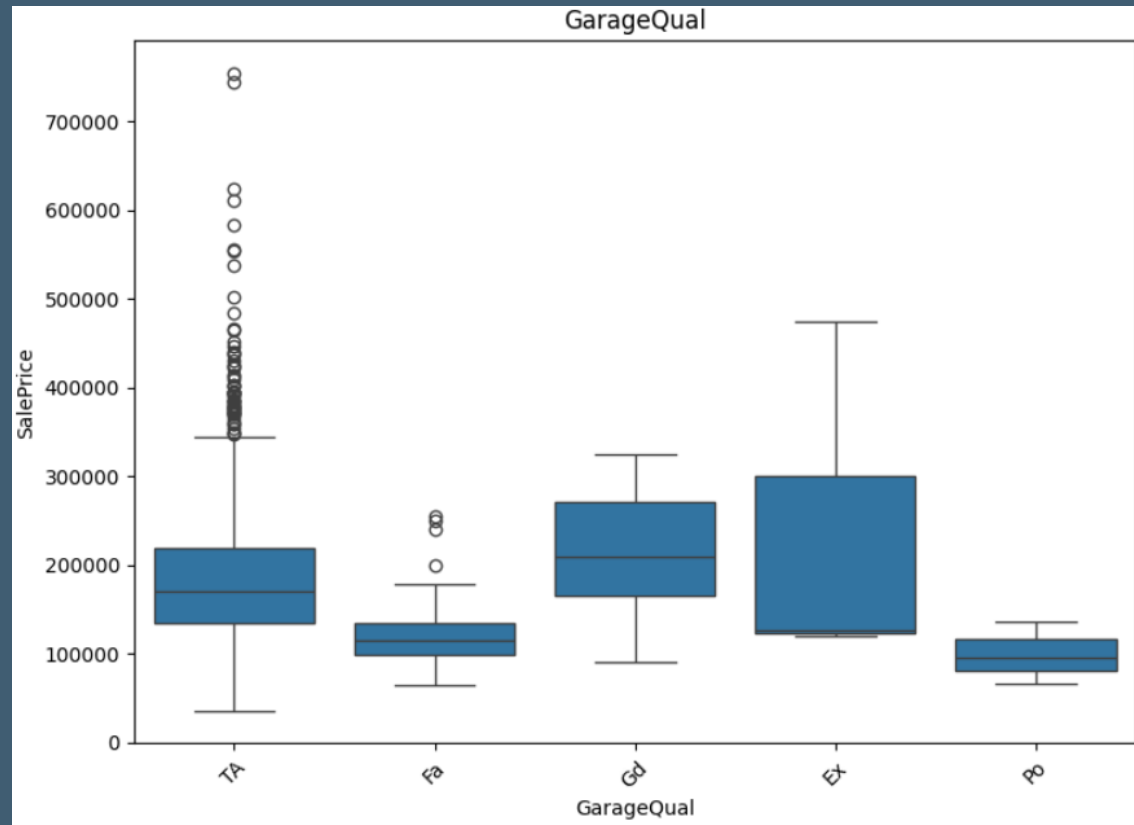
- Basement quality strongly affects price.
- Larger basements increase property value.

Living Area

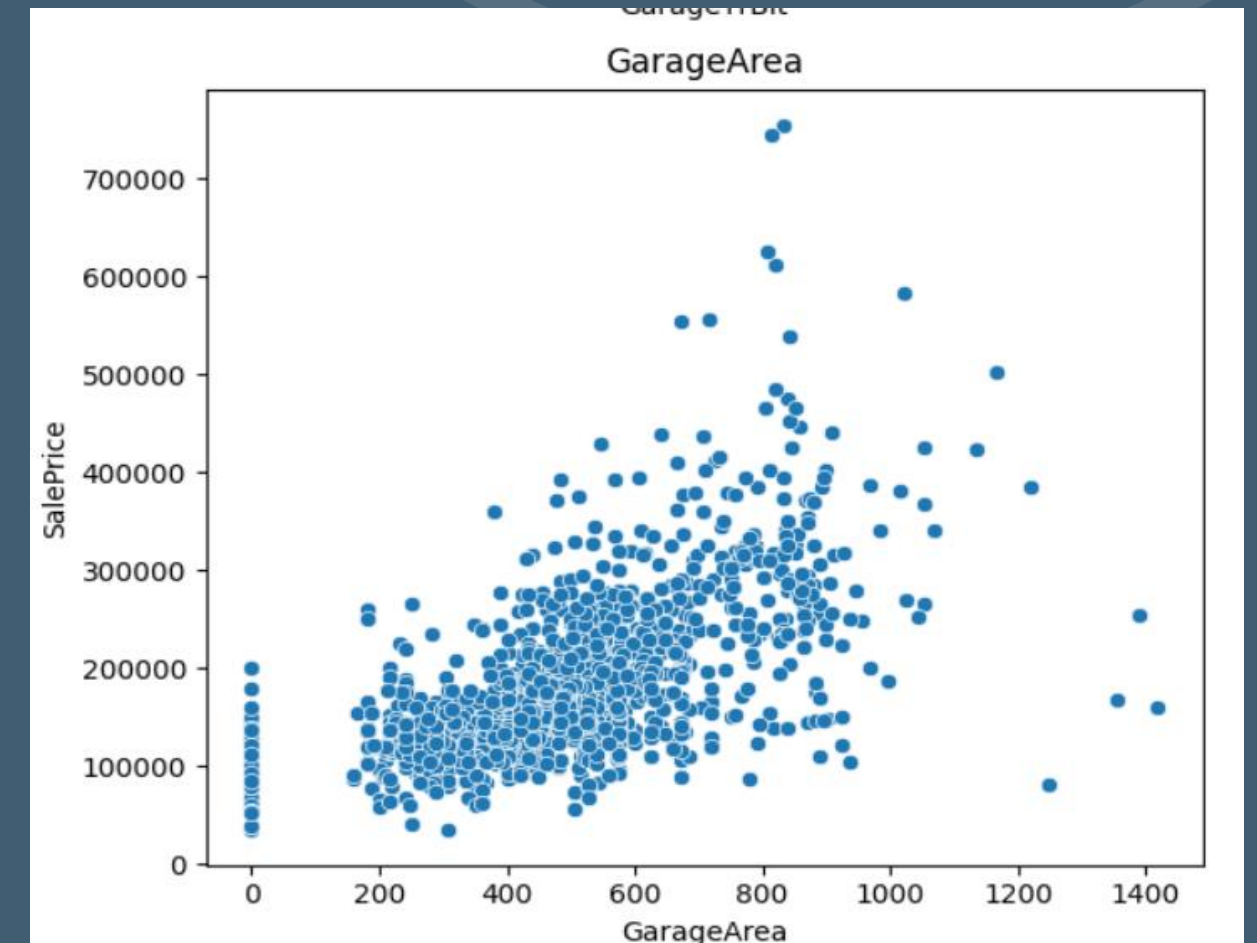
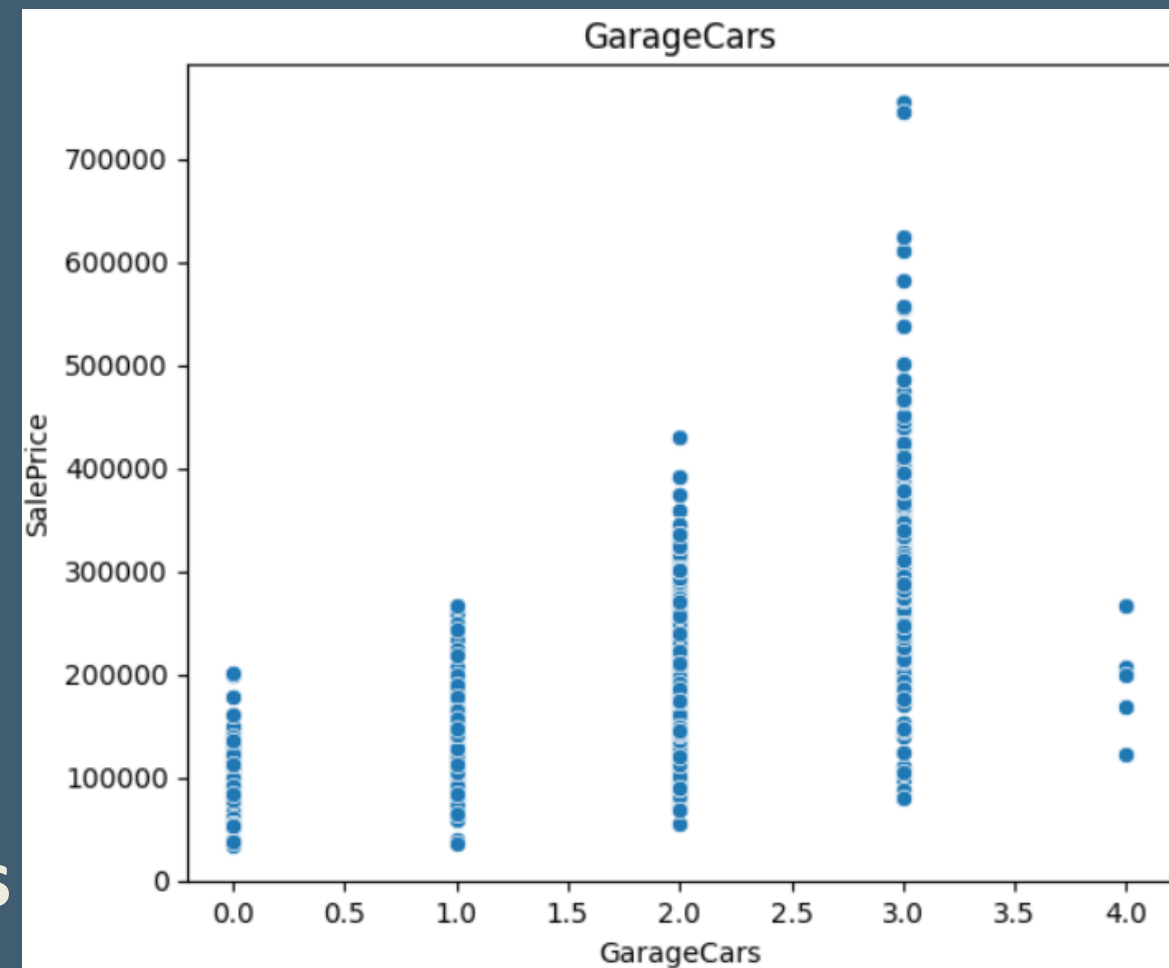


- GrLivArea Correlation = 0.71
- Living area is one of the strongest predictors.

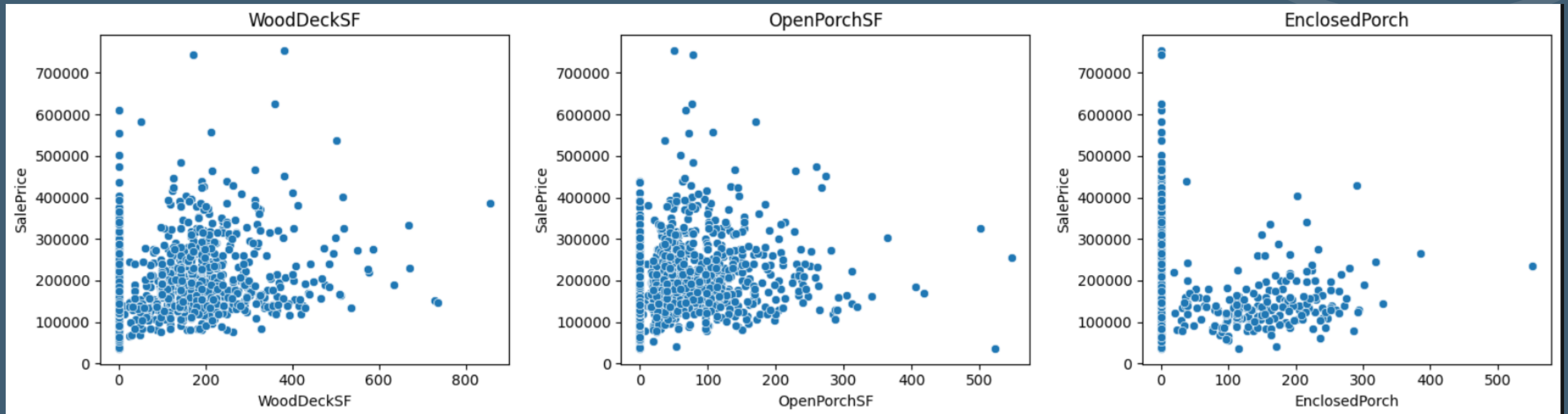
Garage



- Bigger garages increase price.
- Garage quality positively affects value.

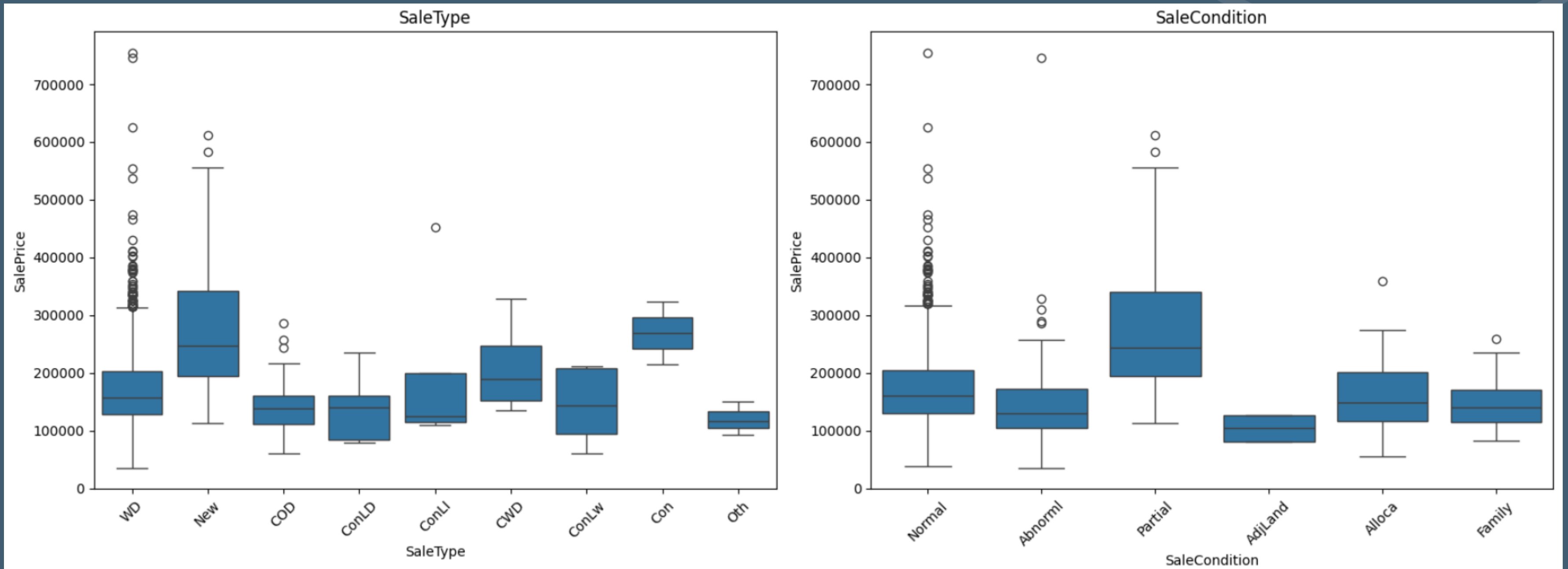


Outdoor Features



- Deck and porch areas moderately increase house prices.
- Pool quality is important but has very few observations.

Sale Information



- Newly constructed houses sold under partial conditions have the highest prices.

Key Findings

Strong Features	Moderate	Weak
OverallQual	WoodDeckSF	MiscVal
GrLivArea	OpenPorchSF	3SsnPorch
TotalBsmtSF	Fireplaces	BsmtFinSF2
GarageCars	FullBath	MoSold

Data Preparation

Missing Values

- Encode "None"
- Median Imputation

Encoding

- One-Hot Encoding
- Ordinal Encoding

Scaling

- StandardScaler

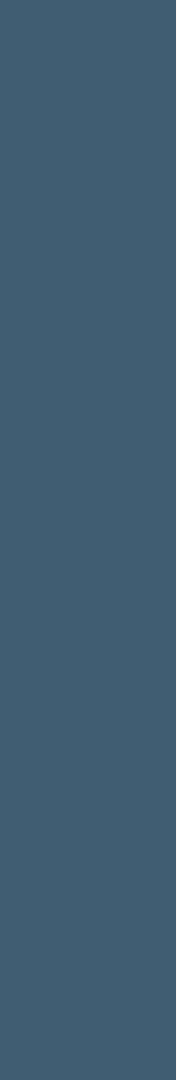
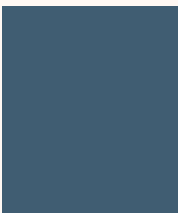
Feature Engineering

- Total Area
- House Age
- Total Bathrooms

Conclusion

Main Findings

- Overall quality is the strongest predictor.
- Living area significantly impacts house prices.
- Basement and garage characteristics contribute positively.
- Neighborhood plays a major role.
- Outdoor features have moderate influence.
- Some variables contain missing values representing the absence of a feature rather than missing data.



THANK YOU

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